

Harnessing β -chitin acetyl-amide chemistry for the fabrication of robust, lightweight, and sustainable aqueous zinc-ion battery separators via centrifugal jet spinning

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ABSTRACT

The development of robust and sustainable separators is paramount for the practical implementation of aqueous Zn-ion batteries (AZIBs). Herein, we report a high-performance β -chitin nanofiber separator (β -ChNF-S) designed to address the chronic issues of Zn dendrite growth and parasitic side reactions. Leveraging the unique acetyl-amide chemistry of β -chitin, the β -ChNF-S separator, fabricated via a scalable centrifugal jet spinning process, acts as an active interfacial medium rather than a passive barrier. Our findings demonstrate that the β -ChNF-S separator effectively perturbs the $[\text{Zn}(\text{H}_2\text{O})_6]^{2+}$ solvation shell, improving interfacial ion-transfer behavior and homogenizing Zn^{2+} flux. In addition, the highly wettable chitin network regulates the interfacial water hydrogen-bonding environment and stabilizes the Zn/electrolyte interface, thereby suppressing parasitic byproduct formation. Consequently, an AZIB cell prepared using the β -ChNF-S exhibits an exceptional cycling stability of over 900 h at high-current densities with a dendrite-free morphology. This work provides a new strategy for utilizing biomass-derived polymers to engineer sophisticated interfaces for next-generation, eco-friendly energy storage systems.

1. Introduction

Aqueous Zn-ion batteries (AZIBs) are increasingly recognized as a promising technology for large-scale energy storage owing to their intrinsic safety, economic feasibility, and environmentally benign nature (Cheng et al., 2023; B. Li et al., 2024; S. Li & Zhi, 2025; H. Ma et al., 2025; Zhao et al., 2024). However, the widespread application of AZIBs remains limited by the critical challenge of Zn dendrite formation and undesired hydrogen evolution reaction (HER) at the Zn anode interface, which severely degrade battery life and stability (Chazalviel, 1990; B. Li

et al., 2024; H. Ma et al., 2025; Zhao et al., 2024). Conventionally, glass fiber separators (GF-S) are utilized in AZIBs because of their affordability and favorable electrolyte wettability (B. Li et al., 2024). However, commercial glass-fiber separators (GF-S) are inherently bulky and heavy (e.g., Whatman GF/B with a nominal thickness of 675 μm and a basis weight of 143 g m^{-2}) and typically possess coarse, heterogeneous pore structures and show weak chemical affinity toward Zn^{2+} , which collectively leads to nonuniform Zn deposition and accelerated dendrite propagation. (Jana et al., 2015; B. Li et al., 2024).

Consequently, there is an urgent need for alternative separators that

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reduce thickness and mass to improve practical Wh kg^{-1} and Wh L^{-1} , while homogenizing Zn^{2+} flux to mitigate dendrite formation and meeting the stringent mechanical and chemical requirements for practical AZIB applications. (B. Li et al., 2024; H. Ma et al., 2025; Zhao et al., 2024). Biomass-derived separators, especially cellulose-based membranes, have emerged as sustainable and eco-friendly alternatives to GF-S owing to their renewability, biodegradability, and natural abundance of hydroxyl functional groups, which promote electrolyte uptake and ionic transport (Cheng et al., 2023; Du et al., 2025; Fu et al., 2022; Xia et al., 2024; W. Yang et al., 2025; H. Zhou et al., 2025; W. Zhou et al., 2022). Cellulose-based separators have emerged as promising sustainable alternatives to conventional glass-fiber separators owing to their renewability, biodegradability, and hydrophilic surface chemistry, which promote electrolyte uptake and ionic transport (Cheng et al., 2023; S. Yang et al., 2023; W. Zhou et al., 2022). Recent engineered cellulose systems, including functionalized nanocellulose and hybrid cellulose-based membranes incorporating MOFs, MXenes, or graphene derivatives, have further demonstrated improved dendrite suppression and electrochemical stability by enhancing dimensional stability and interfacial ion regulation (An et al., 2022; Hong et al., 2020; Y. Li et al., 2023; S. Liu, Hu, et al., 2025; Wang et al., 2024; S. Yang et al., 2023; Z. Zheng et al., 2023). Despite this progress, the simultaneous realization of stable ion regulation, lightweight separator architecture, and simple fabrication remains an important design challenge for practical AZIB separators.

To address these limitations, we utilize β -chitin, which exhibits intrinsic chemical advantages over cellulose due to its unique acetyl-amide and hydroxyl groups (Fan et al., 2010; Weitz et al., 2008). Compared with α -chitin, β -chitin possesses weaker intersheet hydrogen bonding and a less compact crystal packing, leading to higher swelling ability and solvent accessibility. These features can expose hydroxyl and acetyl-amide functionalities more effectively and are advantageous for electrolyte infiltration and ion-accessible separator architectures (Cuong et al., 2016). In this study, we present an innovative and scalable strategy for the fabrication of a β -chitin nanofiber separator (β -ChNF-S) specifically tailored for AZIB applications (Golecki et al., 2014; Jung et al., 2016; Weitz et al., 2008). β -ChNF-S is prepared via centrifugal jet spinning (CJS), a high-throughput and easily scalable production method that offers unique advantages over conventional electrospinning or vacuum filtration processes, which require high-voltage setups, complex solvent management, and longer production times (Golecki et al., 2014; Jung et al., 2016; Weitz et al., 2008). Recent comparative analyses have shown that centrifugal spinning can achieve production rates of $1\text{--}50\text{ g h}^{-1}$, substantially higher than conventional electrospinning ($1\text{--}100\text{ mg h}^{-1}$), while also being compatible with multi-nozzle scale-up manufacturing designs (Huang et al., 2019). In our laboratory-scale CJS setup, the collected dry β -ChNF-S reached a production rate of approximately 0.5 g min^{-1} , corresponding to 30 g h^{-1} . Although HFIP recovery and solvent management remain necessary for large-scale implementation, CJS does not require a high-voltage electric field and offers a simple centrifugal-force-driven route for scalable fibrous separator fabrication. In addition, squid pens are underutilized seafood by-products that can serve as a sustainable β -chitin source, while the thin and lightweight β -ChNF-S architecture reduces inactive separator mass compared with conventional GF-S, supporting its practical potential for high-energy-density AZIBs. The produced β -ChNF-S exhibits a unique combination of high porosity, uniform nanopore distribution, and superior mechanical flexibility, while also providing a thin, lightweight, and compact architecture that minimizes the inactive separator contribution in AZIBs (e.g., $142\text{ }\mu\text{m}$ in this work). While both systems reach a comparable energy density of 364 Wh kg^{-1} excluding the separator mass, normalized evaluations reveal a stark contrast in practical performance. Specifically, the β -ChNF-S based cell delivers approximately 80 Wh kg^{-1} , more than double the 35 Wh kg^{-1} achieved by conventional GF-S. This significant improvement in both gravimetric and volumetric energy density is driven by the reduced thickness, lighter weight, and

superior electrolyte wettability of the β -ChNF-S (Cheng et al., 2023; Fu et al., 2022; Wang et al., 2024; S. Yang et al., 2023; W. Yang et al., 2025; W. Zhou et al., 2022). Moreover, the inherent acetyl-amide and hydroxyl functionalities in β -chitin provide polar and hydrogen-bonding sites that can regulate the local electrolyte environment and guide Zn^{2+} flux without strongly trapping Zn^{2+} ions. Rather than acting as strong Zn^{2+} chelation sites, these functional groups promote weak and reversible interactions with hydrated Zn species and water molecules, which is favorable for maintaining facile ion transport and homogeneous Zn deposition. This interaction is expected to homogenize Zn^{2+} flux at the electrode interface and suppress localized ion accumulation that triggers dendritic Zn growth in aqueous Zn-ion batteries (Hong et al., 2020; Y. Li et al., 2023; C. Liu, Lin, et al., 2025; Wang et al., 2024; W. Yang et al., 2025). β -ChNF-S further demonstrates exceptional mechanical resilience and electrolyte retention under harsh operating conditions, resulting in prolonged cycle stability and enhanced safety performance (B. Li et al., 2024; H. Ma et al., 2025; Monroe & Newman, 2003; Zhao et al., 2024). Therefore, we hypothesize that the acetyl-amide and hydroxyl functionalities of β -chitin provide a balanced and site-dependent interaction environment for hydrated Zn species. While oxygen-rich sites can favorably interact with Zn-containing species, relatively weak adsorption at other exposed surface regions may prevent excessive Zn^{2+} immobilization. This combination of favorable local affinity and weak reversible surface interaction can help retain electrolyte, guide Zn^{2+} migration, and homogenize Zn^{2+} flux at the electrode interface, thereby suppressing localized ion accumulation and dendritic Zn growth in aqueous Zn-ion batteries (Camci-Unal & Pohl, 2010).

This study provides new insights into the utilization of naturally occurring carbohydrate polymers in advanced battery technology. In addition, it underscores the potential of the CJS process for the scalable production of sustainable, high-performance separators, paving the way for the practical and commercial deployment of robust, dendrite-free AZIBs.

2. Materials and methods

2.1. Materials

2.1.1. Materials for the fabrication of β -ChNF-S

β -Chitin was purchased from iCell (Republic of Korea). Sodium hydroxide (NaOH, 98.0%), used for the deproteinization process, was obtained from SAMCHUN (Republic of Korea). 1,1,1,3,3,3-Hexafluoro-2-propanol (HFIP), used as a solvent to dissolve β -chitin, was purchased from SEMK Corporation (Republic of Korea).

2.1.2. Battery components and electrolytes

Vanadium pentoxide (V_2O_5), obtained from Junsei Chemical (Japan), was used for cathode preparation. Conductive carbon black and polyvinylidene fluoride binder were purchased from Thermo Fisher Scientific (USA) and Sigma-Aldrich (USA), respectively. *N*-Methyl-2-pyrrolidone (Sigma-Aldrich) was used as the solvent for the cathode slurry. Zn foil (0.25 mm thick, 99.9%) was purchased from Thermo Fisher Scientific (USA) and used as the anode. To prepare the electrolyte, Zn trifluoromethanesulfonate ($\text{Zn}(\text{CF}_3\text{SO}_3)_2$, 98%; Sigma-Aldrich) was dissolved in deionized (DI) water. Commercial glass fiber (GF/F, Whatman) membranes were used as reference separators. All chemical reagents were used as received without further purification.

2.2. Preparation of β -ChNF-S

Commercial β -chitin was pretreated via cryomilling before alkaline deproteinization according to our previous work (Kim et al., 2024). Briefly, β -chitin granules were loaded into a milling jar together with four zirconia balls and processed using a cryomilling machine (Retsch, Germany) for 60 s to obtain finely ground powders. After removing the zirconia balls, 4 g of the cryomilled β -chitin powder was dispersed in 50

mL of 4 wt% NaOH aqueous solution and heated in an oven at 90 °C for 2 h. The resulting suspension was then thoroughly washed with DI water until the filtrate reached a neutral pH. This alkaline deproteinization/washing procedure was repeated three times. Then, the deproteinized β -chitin was completely dried in an oven at 60 °C for 24 h, and the dried agglomerates were finally pulverized using a kitchen blender to obtain a homogeneous powder. The purified β -chitin has a high degree of acetylation ($\sim 97\%$), crystallinity ($\sim 82\%$), and high purity with protein/ash residue of less than 0.2 wt% and 0.03 wt%, respectively, as characterized in our previous literature (Table S1) (Kim et al., 2024). The β -chitin powder was then dissolved in HFIP (0.8 w/v%) to prepare a spinning dope (50 mL). Subsequently, the dope was subjected to CJS using a dual nozzle (30 G) at 5500 rpm, producing dry-spun β -ChNF-S fibers (Jeong et al., 2018; Kim et al., 2017). For the preparation of the separators, the as-spun fibers were collected, compressed using a roll press to form a dense sheet, and then punched into circular disks ($\Phi 19$ mm) for subsequent use.

2.3. Characterization

The fiber morphology was examined using field-emission scanning electron microscopy (Cold FE-SEM, Hitachi SU8220, Japan). The hierarchical pore structure and distribution of the β -ChNF-S fibers were thoroughly investigated using mercury intrusion porosimetry (MIP; AutoPore IV 9500, Micromeritics, USA) and capillary flow porometry (CFP; CFP-1500AEX, Porous Materials Inc. (PMI), USA). Gurley air permeability was measured using a Gurley densometer (Model 4118 N, Gurley Precision Instruments, USA), and membrane porosity was calculated from weight and thickness measurements. Gurley air permeability and electrolyte uptake were performed using three independent separator samples for each condition, and the results are presented as mean $\pm$ standard deviation. Crystallinity and the β -polymorphic phase of chitin were analyzed via X-ray diffraction (XRD; Mini Flex 600, Rigaku, Japan) with Cu K α radiation ($\lambda = 1.5406$ Å). Fourier transform infrared (FTIR) spectroscopy (Nicolet iS50, Thermo Fisher, USA) was conducted in the range of 4000–800 cm^{-1} to monitor acetyl-amide and hydroxyl vibrations and evaluate the hydrogen-bonding environment of electrolyte-wetted separators. Thermal resistance was examined by placing the membranes on a hot plate at 100 °C, 150 °C, and 200 °C for 30 min and by performing thermogravimetric analysis –differential thermogravimetric (TGA-DTG) analysis (TGA/SDTA 851e, Mettler Toledo (Swiss)) from 30 °C to 1000 °C at a heating rate of 10 °C min^{-1} under N_2 . Confocal laser scanning microscopy (CLSM; Carl Zeiss (LSM 800 MAT)) was employed to evaluate the surface roughness of the Zn electrode after cycling with different separators.

2.4. Computational details (DFT)

Density functional theory (DFT) calculations were performed using the Vienna ab initio simulations package (VASP) to investigate the interaction behavior of Zn atom with β -chitin (Hohenberg & Kohn, 1964). The projector-augmented wave (PAW) method and a plane-wave basis set with a cut-off energy of 520 eV were employed to describe the core-electron contributions (Kresse & Furthmüller, 1996; Kresse & Joubert, 1999). The exchange-correlation functional was treated within the generalized gradient approximations (GGA) using Perdew-Burke-Ernzerhof (PBE) (Perdew et al., 1996). To account for van der Waals interactions, the DFT-D3 method of Grimme with Becke-Johnson damping was employed (Grimme et al., 2010, 2011). The bulk β -chitin structure was adopted from Nishiyama et al. (Nishiyama et al., 2011). The Brillouin zone was sampled using a Monkhorst-Pack k-points mesh of $4 \times 2 \times 2$. The total energy convergence criterion was set to 10^{-5} eV, and ionic relaxation was performed with tolerance of 0.03 eV Å $^{-1}$. For β -chitin surface, a slab model consisting of a single layer was constructed. A Γ -centered k-point grid of $5 \times 1 \times 2$ was used, with an electronic convergence threshold of 10^{-5} eV and a force

convergence criterion of 0.05 eV Å $^{-1}$.

2.5. Evaluation of the electrochemical performance of AZIBs

Electrochemical tests were performed in a half-cell configuration using CR2032 coin cells and a 3 M $\text{Zn}(\text{CF}_3\text{SO}_3)_2$ aqueous solution in DI water as the electrolyte. $\text{Zn}||\text{Zn}$ symmetric cells were assembled using two Zn foils as both electrodes separated by either GF-S or β -ChNF-S. Galvanostatic cycling was performed on a battery cyclers (NEWARE, China). $\text{V}_2\text{O}_5||\text{Zn}$ cells were fabricated using Zn foil as the anode and commercial V_2O_5 cathodes (1 mg cm^{-2} loading). Charge-discharge tests were conducted at various current rates (0.1–2 A g^{-1}) within a voltage window of 0.4–1.8 V. Long-term cycling stability was examined for >1000 cycles at 1 and 2 A g^{-1} . Specific energy values (Wh kg^{-1}) were normalized with and without considering the separator weight. Electrochemical impedance spectroscopy (EIS; BioLogic SP-200) was performed in the frequency range of 1 MHz–0.1 Hz with an amplitude of 10 mV.

3. Results and discussion

3.1. Fabrication and structural properties of β -ChNF-S

As illustrated in Fig. 1a, β -chitin derived from squid pens was utilized as a sustainable carbohydrate polymer precursor for separator fabrication. Amide-bearing chitin/chitosan matrices have been explored as ion-regulating media in aqueous Zn systems and, more broadly, as sustainable separators for batteries (García-Castrillo et al., 2026; Song et al., 2024). Squid pens are an abundant yet underutilized biomass resource that can be efficiently processed to yield β -chitin, which is characterized by a highly ordered crystalline structure in its β -polymorphic form (Cuong et al., 2016; Q. Ma et al., 2019). The β -ChNF-S membrane was fabricated using the CJS method, which enables the high-throughput, efficient, and scalable manufacturing of large-area fibrous membranes (Fig. 1b) (Golecki et al., 2014; Weitz et al., 2008). Unlike conventional electrospinning, which requires high-voltage equipment and is constrained by solvent compatibility and production rates, CJS facilitates the rapid, continuous formation of uniform nanofibers without requiring a high-voltage electric field. These inherent advantages render CJS particularly attractive for industrial-scale production of separator membranes while minimizing environmental impact. The resulting β -ChNF-S membranes exhibit excellent membrane-forming characteristics and mechanical robustness. Tensile stress-strain curves (Fig. S1) further demonstrated that β -ChNF-S possesses a tensile strength of 26.9 MPa and a Young's modulus of 1.2 GPa, supporting its reliable handling and structural integrity as a separator. Such flexible and robust characteristics are advantageous over conventional GF-S, which is generally brittle and mechanically less adaptable during cell assembly. At the molecular scale, β -chitin uniquely possesses acetyl-amide and hydroxyl functional groups, which can provide polar interaction and hydrogen-bonding sites for hydrated Zn species and water molecules. As schematically depicted in Fig. 1c, these functional groups are expected to weakly and reversibly interact with Zn species rather than strongly chelating or trapping Zn^{2+} ions. Such moderated surface interactions can assist Zn^{2+} flux regulation and interfacial water-structure modulation, thereby reducing local ion concentration gradients and current-density fluctuations at the Zn–electrolyte interface. In addition, the hydrogen-bonding capability of the β -chitin framework may decrease the availability of weakly bound/free water near the electrode surface, contributing to the suppression of undesired HER during electrochemical cycling. As revealed in the SEM images (Fig. 1d,e), the separator comprises a randomly interconnected network of nanofibers with an average diameter of approximately 0.8 μm (see low- and high-magnified SEM images in Fig. S2). This fibrous architecture confers substantial porosity and interconnected pore channels, which are highly favorable for electrolyte uptake, efficient ionic diffusion, and

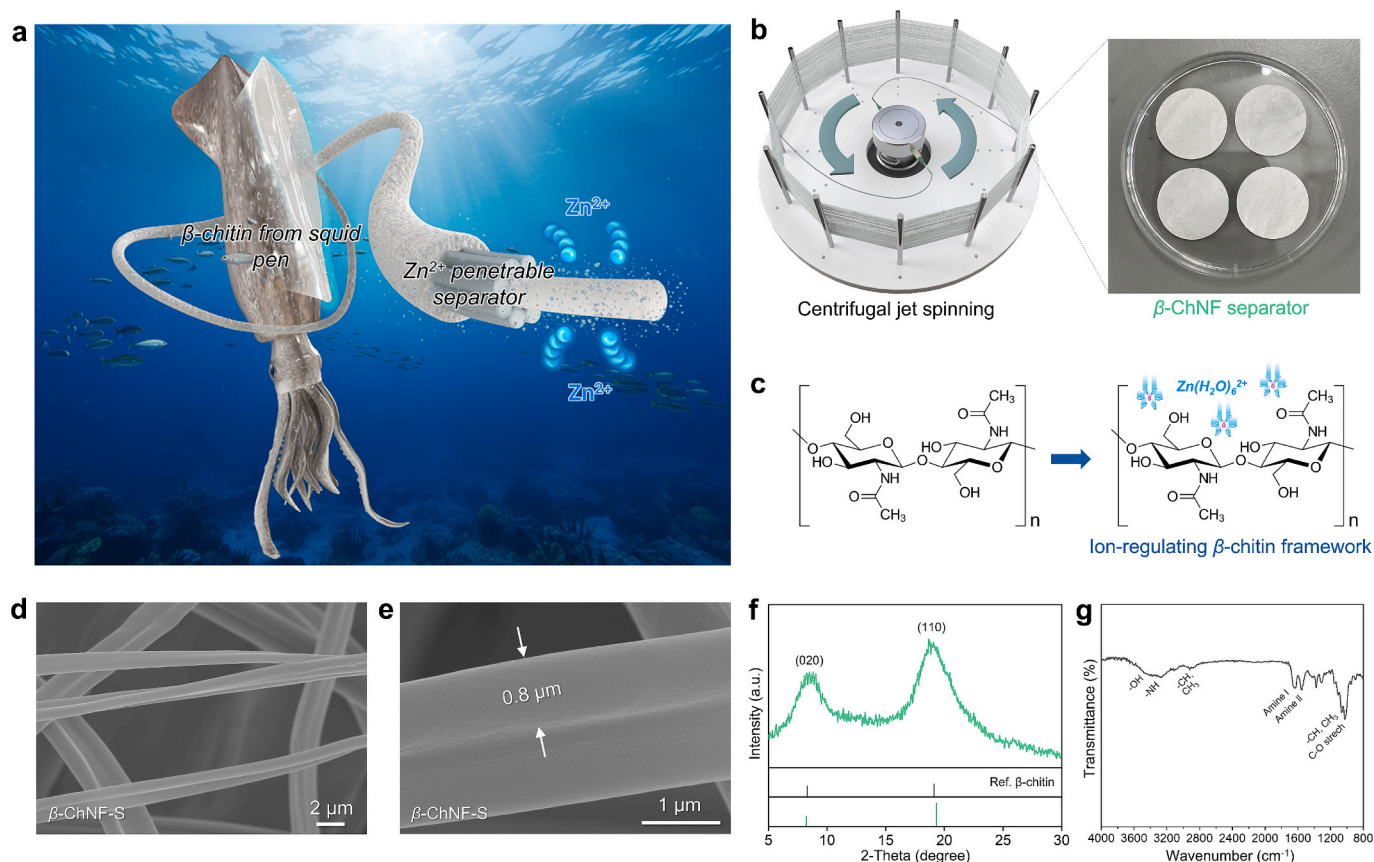


Fig. 1. a) Schematic illustration showing ion-regulating β -chitin fibers derived from squid pen. b) β -ChNF-S manufactured through centrifugal jet-spinning (CJS). c) Structural formula of β -chitin showing hydroxyl/acetyl-amide polar groups and weak/reversible Zn interaction. d,e) SEM image of the CJS-processed β -ChNF-S. f) XRD profile and g) FT-IR spectra of the β -ChNF-S.

mechanical compliance. Furthermore, this nanofibrous morphology enhances dimensional stability in aqueous electrolytes, overcoming a common limitation of cellulose-based separators. FTIR spectroscopy was performed to confirm the chemical structure of β -chitin in the membrane (Fig. 1g). The spectrum exhibited representative vibrational bands of β -chitin, including amide I and amide II bands, verifying that the intrinsic β -chitin backbone structure was well preserved after membrane fabrication. These results confirm that the membrane maintains the inherent chemical structure of β -chitin.

3.2. Electrolyte wettability, thermal stability, and permeability characteristics of β -ChNF-S

The electrolyte wettability of battery separators critically influences ionic transport and interfacial resistance in AZIBs (W. Zhou et al., 2024). As demonstrated in Fig. 2a, β -ChNF-S exhibited superior electrolyte uptake compared to conventional GF-S. In addition, quantitative electrolyte uptake measurements further highlighted the superior wettability of β -ChNF-S (Fig. 2b); β -ChNF-S exhibited an electrolyte uptake of $\sim 3438\%$, substantially higher than that of GF-S ($\sim 1200\%$). This high uptake suggests that β -ChNF-S can act as a stable electrolyte reservoir, helping sustain continuous ionic pathways and mitigating localized ion starvation during cycling. In contrast, GF-S exhibited comparatively sluggish electrolyte absorption kinetics, potentially resulting in heterogeneous Zn^{2+} distribution and nonuniform Zn deposition during cycling. The high electrolyte uptake of β -ChNF-S highlights its strong electrolyte affinity and efficient capillary-driven wettability within the interconnected fibrous network. This highly wettable scaffold enables rapid and homogeneous electrolyte infiltration throughout the separator, establishing continuous ion-conduction pathways even with a limited

electrolyte amount. Such uniform electrolyte distribution can mitigate local ion depletion and concentration polarization during Zn plating/stripping, thereby promoting stable interfacial reactions. From a practical cell-design perspective, efficient electrolyte utilization is advantageous for improving the electrolyte-to-capacity (E/C) balance, as stable ion transport and accessible capacity can be maintained without unnecessary electrolyte excess. The enhanced wettability of β -ChNF-S was confirmed by dynamic contact angle measurements (Fig. 2c), which revealed that the β -ChNF-S membrane displayed an exceptionally rapid reduction in contact angle from approximately 50° to nearly 15° within 0.3 s. This rapid electrolyte absorption underscores the inherently hydrophilic characteristics of β -chitin, which mainly stem from the abundant hydroxyl and acetyl-amide functional groups along the polymer backbone. To substantiate the intrinsic polarity of β -ChNF-S responsible for the rapid wetting behavior, X-ray photoelectron spectroscopy was performed (Fig. S3). The C 1s spectrum showed pronounced contributions from polar bonding environments (e.g., C–O/C–N and carbonyl-related species), accompanied by clear O 1s and N 1s signals, confirming the presence of oxygen- and nitrogen-containing functional groups consistent with hydroxyl- and acetyl-amide bearing moieties of polysaccharides. Such distributed polar functionalities confer strong electrolyte affinity and provide hydrogen-bonding sites for regulating the local water structure. To further examine this hydrogen-bond-mediated water-structure regulation, FT-IR analysis was performed using electrolyte-wetted GF-S and β -ChNF-S separators, with 3 M $\text{Zn}(\text{CF}_3\text{SO}_3)_2$ electrolyte as a reference (Fig. S4). Compared with GF-S/electrolyte, β -ChNF-S/electrolyte exhibited a red-shift of the O–H stretching maximum from 3366.72 to 3262.35 cm^{-1} , indicating strengthened hydrogen bonding between water molecules and the hydroxyl-rich β -ChNF-S framework. Deconvolution of the O–H stretching region

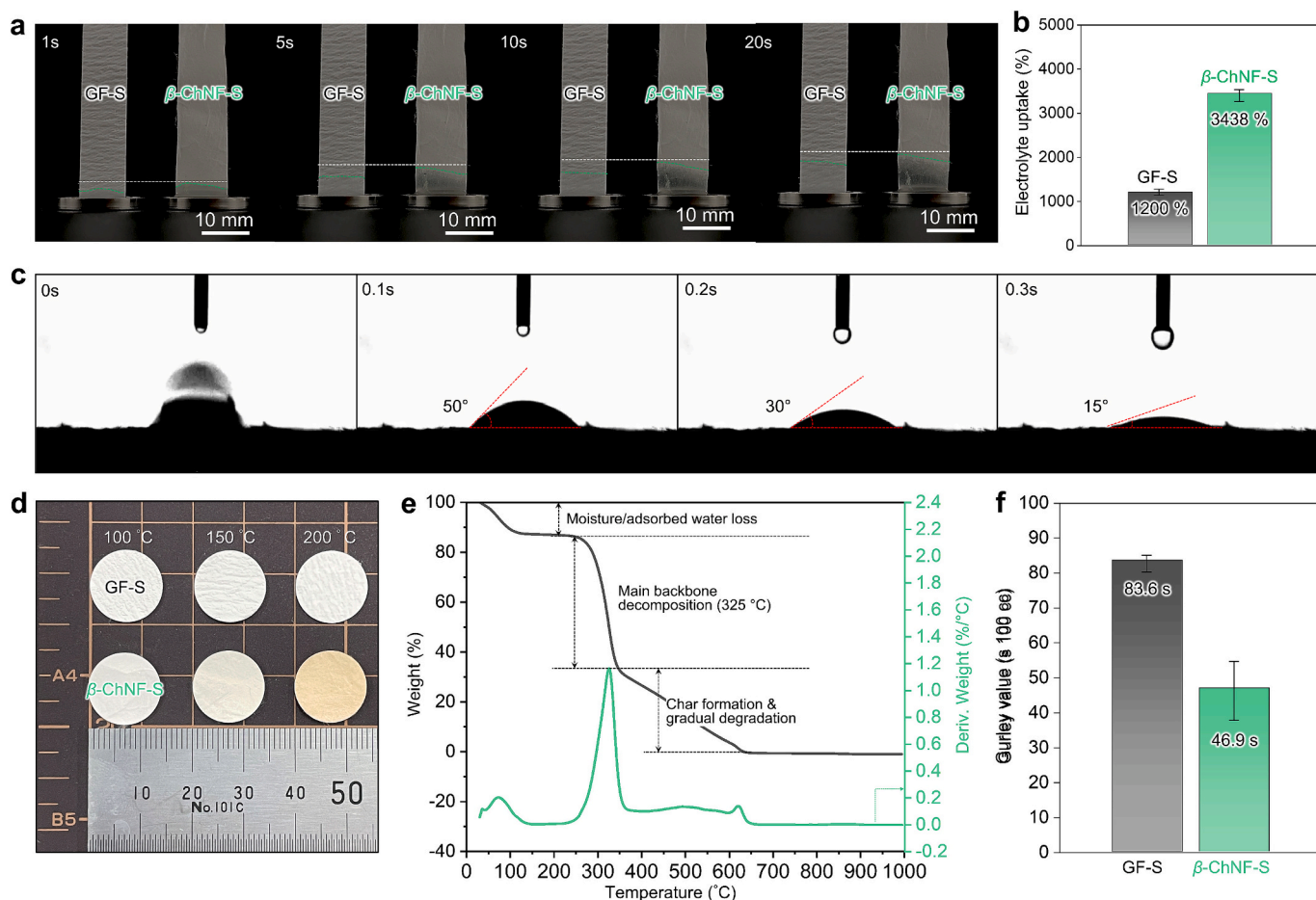


Fig. 2. a) Comparison of electrolyte wettability of a glass fiber separator (GF-S) and the β -chitin nanofiber separator (β -ChNF-S). b) Electrolyte uptake test of GF-S and β -ChNF-S. c) Contact angle test of β -ChNF-S. d) Thermal stability test of GF-S and β -ChNF-S at high temperatures (100 °C, 150 °C, 200 °C). e) TGA profiles of β -ChNF-S. f) Gurley value analysis.

further showed that the strong hydrogen-bonded water fraction increased from 28.19% to 33.79%, while the weak/free water fraction decreased from 21.83% to 17.08%. These results suggest that β -ChNF-S regulates the interfacial solvation environment by reconstructing the water hydrogen-bonding network and reducing weak/free water, rather than by strongly adsorbing or trapping Zn^{2+} ions. The thermal stability of separators was rigorously evaluated to ensure their structural integrity under realistic operating conditions. As illustrated in Fig. 2d, the thermal robustness of the β -ChNF-S membranes was evaluated by examining their macroscopic appearance upon heating at 100 °C, 150 °C, and 200 °C. GF-S exhibited negligible changes across the tested temperatures, whereas β -ChNF-S retained its disk shape without noticeable shrinkage or deformation even at 200 °C, showing excellent morphological stability. A mild yellowing was observed for β -ChNF-S at 150 °C–200 °C; however, the absence of apparent collapse indicated that the membrane maintained structural integrity at high temperatures, demonstrating thermal performance comparable to, or potentially exceeding that of conventional GF-S. The robust thermal behavior of β -ChNF-S was further examined via TGA–DTG (Fig. 2e). The TGA profile showed an initial minor mass loss at low temperatures, followed by a dominant decomposition in the mid-temperature range and a gradual mass loss at higher temperatures. Consistently, the DTG curve displayed a pronounced maximum at ~300 °C–350 °C, which corresponds to the primary decomposition of the β -ChNF-S backbone, and a subsequent weight loss from ~350 °C to 600 °C due to progressive degradation/carbonization of the residual fraction. Overall, this stepwise degradation behavior supports the thermal stability of β -ChNF-S well beyond the

practical operating window of AZIBs, suggesting that β -ChNF-S can reliably tolerate thermal stress under prolonged cycling and harsh potential conditions. In addition to electrolyte wettability and thermal robustness, gas permeability is another critical parameter influencing separator performance. Gurley value analysis (Fig. 2f) showed that β -ChNF-S exhibited considerably lower air resistance than GF-S, indicating enhanced porosity and permeability. Complementary porosimetry analyses clarified the pore-level origin of the favorable permeability. CFP revealed a dominant pore-size population centered at ~3 μm , with an appreciable distribution extending toward ~5 μm (Fig. S5), consistent with facile electrolyte infiltration and low air resistance. MIP additionally indicated an accessible porosity of ~31.92% for β -ChNF-S (Fig. S6), supporting the presence of interconnected open-pore structures. Taken together, these results corroborate that β -ChNF-S provides a well-percolated transport network for rapid electrolyte penetration while maintaining sufficient structural stability. Such characteristics directly facilitate efficient electrolyte infiltration and rapid Zn^{2+} diffusion across the separator. Importantly, the optimized pore architecture of β -ChNF-S not only supports elevated ionic conductivity but also maintains adequate mechanical strength, effectively mitigating dendritic growth and penetration.

3.3. Suppression of Zn dendrite formation enabled by β -ChNF-S

To directly evaluate the efficacy of β -ChNF-S in suppressing Zn dendrite formation, $\text{Zn}||\text{Zn}$ symmetric cells were cycled and subsequently disassembled after the fifth charging step. The resulting Zn

electrode surfaces were systematically characterized using SEM and CLSM (Fig. 3). In stark contrast with the β -ChNF-S-based electrodes, Zn electrodes using conventional GF-S exhibited extensive, uncontrolled dendritic growth. The SEM images of the latter electrodes (Fig. 3a,b) clearly showed prominent needle-like Zn dendrites arranged vertically across the electrode surface, indicating severely localized current densities and heterogeneous Zn^{2+} flux. Such irregular deposition patterns typically accelerate dendrite initiation and propagation, thereby reducing electrochemical stability and increasing considerably the risk of internal short circuiting (Guo et al., 2023; Hu et al., 2025). Conversely, the Zn surface retrieved from β -ChNF-S-based cells exhibited a notably smoother and more uniform morphology under identical electrochemical cycling conditions. The corresponding SEM images (Fig. 3c,d) showed a dense, homogeneous Zn deposit layer devoid of dendritic structures. These observations strongly suggest that β -ChNF-S effectively regulates Zn^{2+} flux, suppresses dendrite nucleation, and facilitates uniform ion deposition. Such performance improvements likely arise from molecular-level interactions between Zn^{2+} and the abundant hydroxyl and acetyl-amide functional groups in β -chitin, which enhance desolvation kinetics and enable uniform charge transfer across the Zn–electrolyte interface. Surface topology analyses conducted using CLSM confirmed the SEM results. The three-dimensional CLSM topography of the GF-S–based electrode surface (Fig. 3e) revealed a highly rough and heterogeneous morphology, with height variations exceeding 30 μm , indicative of pronounced dendrite growth. In contrast, β -ChNF-S resulted in a Zn electrode surface (Fig. 3f) with a markedly flatter, smoother profile, exhibiting minimal height fluctuations. This substantial reduction in surface roughness directly correlates with the dendrite-inhibiting properties of β -ChNF-S, mediated by interfacial water-structure regulation and homogeneous Zn^{2+} flux. Although β -ChNF-S possesses micron-scale pores, its dendrite-suppressing behavior should not be interpreted as simple nanoscale pore-size exclusion. Instead, the highly wettable and interconnected fibrous network promotes uniform electrolyte distribution and mitigates localized Zn^{2+} depletion/current-density hotspots, thereby inducing compact and uniform Zn deposition. Beyond morphological stabilization, the chemical by-product formation on the cycled Zn surface was further examined by ex-situ XRD analysis. After 500 cycles in $\text{V}_2\text{O}_5||\text{Zn}$ full cells, the GF-S-based Zn anode exhibited distinct diffraction peaks assigned to zinc hydroxide triflate-type by-products, $\text{Zn}_x(\text{OH})_y(\text{CF}_3\text{SO}_3)_{2x-y}\cdot n\text{H}_2\text{O}$, whereas these

characteristic peaks were markedly weakened for the β -ChNF-S-based Zn anode (Fig. S7). This result supports that β -ChNF-S suppresses the accumulation of passivating by-products during cycling, consistent with the stabilized Zn/electrolyte interface and homogeneous Zn^{2+} flux. The separator stability after cycling was further verified by ex-situ SEM and FT-IR analyses. After 500 cycles in $\text{V}_2\text{O}_5||\text{Zn}$ full cells, β -ChNF-S retained its interconnected fibrous morphology without severe collapse, and its characteristic β -chitin FT-IR bands were largely preserved (Fig. S8 and Fig. S9), confirming the structural and chemical stability of β -ChNF-S in the 3 M $\text{Zn}(\text{CF}_3\text{SO}_3)_2$ electrolyte.

3.4. Enhanced electrochemical stability and dendrite suppression in symmetric cells with β -ChNF-S

Fig. 4a illustrates the galvanostatic cycling stability of $\text{Zn}||\text{Zn}$ symmetric cells at a high current density of 10 mA cm^{-2} and an areal capacity of 1 mAh cm^{-2} . The cells employing β -ChNF-S demonstrated exceptional cycling stability for over 900 h with negligible voltage polarization fluctuations. In marked contrast, the reference cells with traditional GF-S exhibited pronounced instability, characterized by escalating voltage polarization and eventual short circuiting after approximately 600 h. To determine the optimal configuration for effective dendrite suppression, the cycling stability was further evaluated using β -ChNF-S of varying thicknesses (50, 120, 142, and 188 μm) (Fig. 4b). While the 50 μm thick membrane underwent premature short circuiting at approximately 377 h, the 142 μm thick separator provided the most stable voltage profile with minimal fluctuations. These results indicate that membrane thickness is a key architectural parameter for β -ChNF-S, as it governs the balance between electrolyte retention, continuous ion transport, and mechanical resistance to dendrite penetration. The significant extension in cycle life afforded by β -ChNF-S is primarily attributed to the uniform ionic flux distribution and effective dendrite suppression facilitated by the polar/hydrogen-bonding network and flux-homogenizing capability of the β -chitin polymer (Kim et al., 2017). This observation is consistent with recent studies on chitosan-based or polyelectrolyte-complex separators, which have also highlighted the potential of amide-bearing polysaccharides to stabilize Zn plating/stripping by regulating interfacial chemistry (Arif et al., 2025; L. Li et al., 2025; Yu et al., 2025). Further electrochemical insights into the performance of β -ChNF-S were obtained by performing

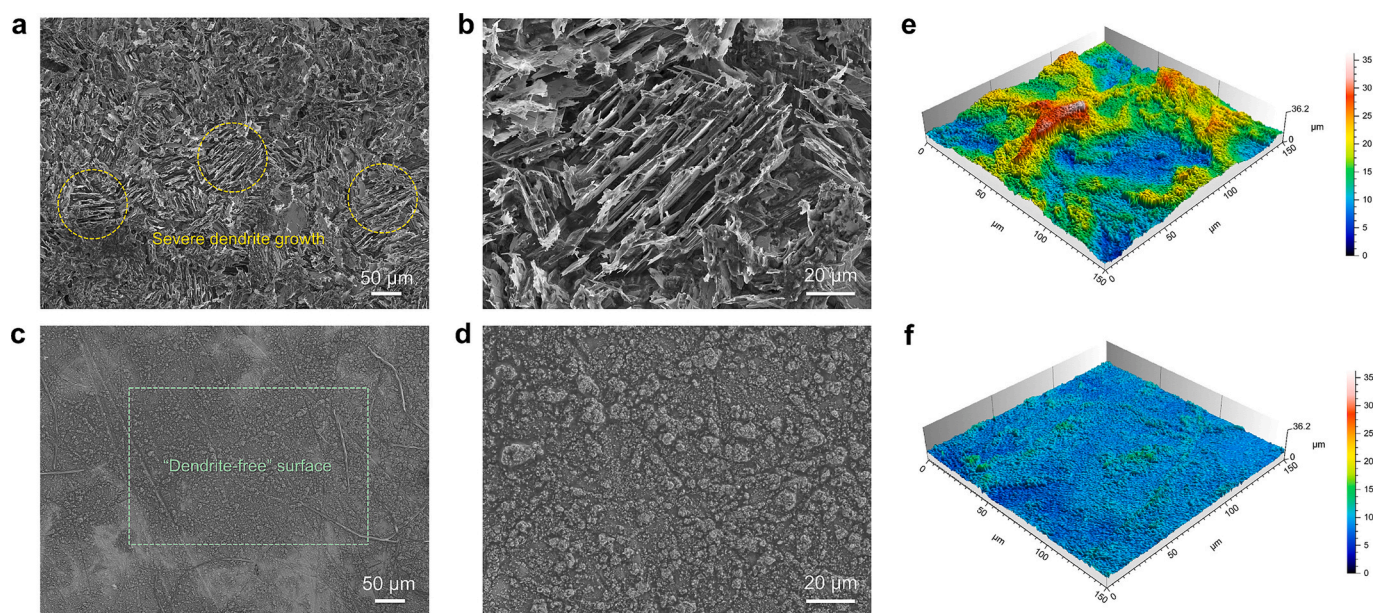


Fig. 3. Characterization of the Zn metal electrode surface retrieved from $\text{Zn}||\text{Zn}$ symmetric cells after the fifth charge. a, b) SEM images of the GF-S-based electrode. c, d) SEM images of the β -ChNF-S-based electrode. e) CLSM images of the GF-S-based electrode. f) CLSM images of the β -ChNF-S-based electrode.

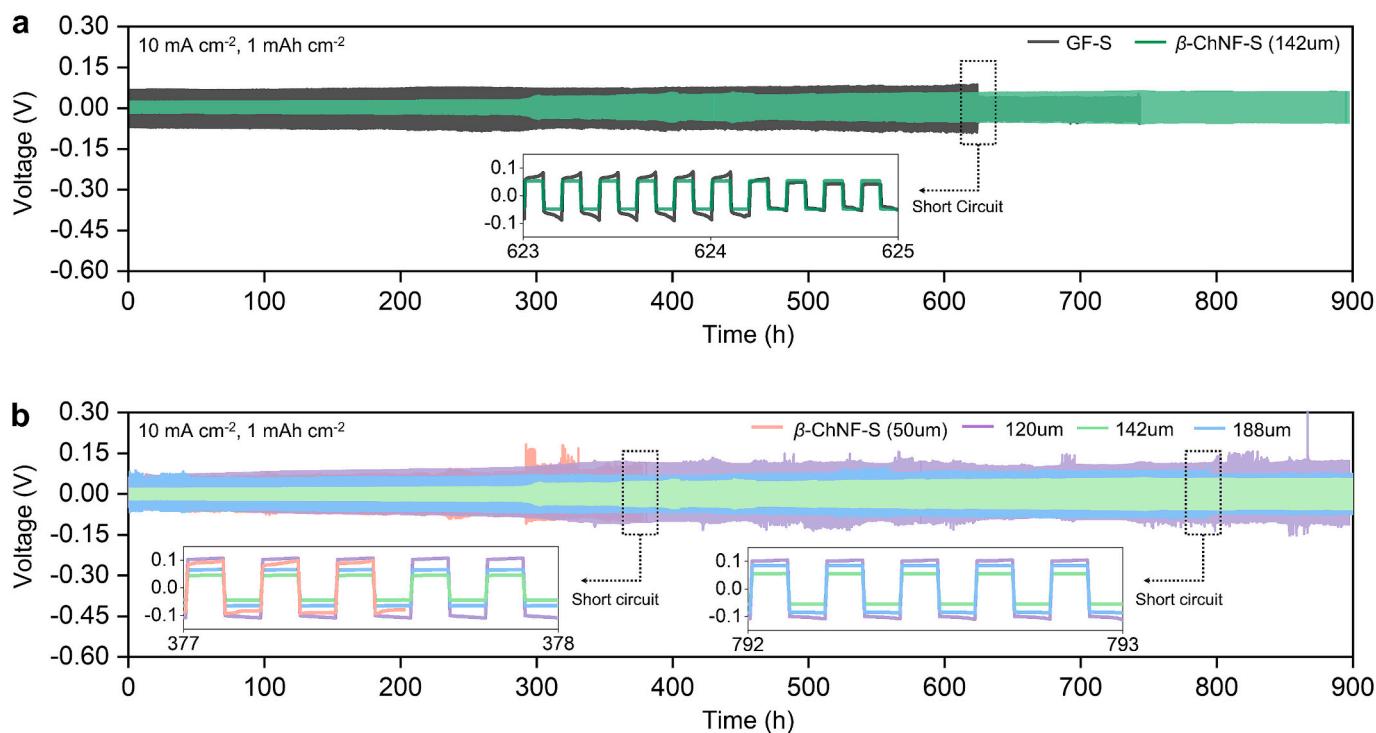


Fig. 4. a) Galvanostatic cycling stability of Zn||Zn symmetric cells assembled with GF-S and β -ChNF-S (142 μm). b) Comparison of β -ChNF-S thickness.

stepwise polarization tests at incrementally increasing current densities ($0.5 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 5 \rightarrow 0.5 \text{ mA cm}^{-2}$), as shown in Fig. S10. Notably, the GF-S-based cell exhibited lower apparent polarization at initial low-current steps, which is consistent with the highly open, low-tortuosity nature of glass fiber matrices with minimized ohmic (iR) contribution. However, at elevated current densities, the GF-S-based cell showed pronounced voltage instability and ultimately underwent short circuit failure, indicating that a lower overpotential does not necessarily translate into a more stable Zn interface under harsh plating/stripping conditions. In contrast, although the β -ChNF-S-based cell exhibited slightly higher polarization in parts of the rate sequence, it maintained stable and continuous cycling without abrupt voltage collapse. Importantly, upon reverting to the initial low-current density (0.5 mA cm^{-2}) after high-current operation, the β -ChNF-S cell rapidly recovered a steady low-polarization profile, demonstrating superior electrochemical reversibility and structural robustness. Overall, these results support that β -ChNF-S provides a more reliable Zn plating/stripping environment under dynamic current loads by suppressing dendrite-induced failure rather than merely lowering the initial iR drop. Potentiodynamic polarization (Tafel) analysis was conducted using Ti||Zn cells to evaluate the effect of the separator on the cell-level polarization behavior relevant to AZIBs (Fig. S11). Compared with the GF-S-based cell, the β -ChNF-S-based cell exhibited a markedly reduced polarization tendency in the cathodic regime, as reflected by a substantially smaller apparent cathodic Tafel slope within the selected fitting region ($\sim 31.6 \text{ mV dec}^{-1}$ for β -ChNF-S vs $\sim 58.8 \text{ mV dec}^{-1}$ for GF-S). This indicates that, under comparable electrochemical driving forces, β -ChNF-S-based cells require a smaller increase in overpotential to sustain a decade increase in current, consistent with a more favorable current-potential response for Zn metal electrode operation. Because the measurements were performed in a two-electrode configuration where separator-induced iR contributions are inherently included, the extracted Tafel parameters are interpreted as apparent values for comparison between separators rather than absolute kinetic constants. Nevertheless, the reduced apparent polarization observed with β -ChNF-S suggests improved electrochemical stability under practical cycling conditions, as corroborated by the galvanostatic polarization and reversibility

results discussed below (Wang et al., 2025; S. Zheng et al., 2025). Related separator engineering strategies—ranging from nanochitin approaches that tune free-water activity and Zn^{2+} solvation (Lin et al., 2025) to conductive/ceramic-modified cellulose nanofiber membranes (Niu et al., 2024) and dielectric interlayers that smooth local electric fields—rely on the same principle of suppressing tip-enhanced deposition, which triggers dendrite growth (Zong et al., 2023). Conversely, GF-S-based cells exhibited pronounced irreversibility and polarization instability, indicating morphological degradation induced by dendritic growth. Taken together, these electrochemical results demonstrate that β -ChNF-S substantially enhances the stability of Zn metal anodes, effectively suppresses dendritic formation, and substantially extends the operational lifetime of AZIBs. Such performance improvements are enabled by the integrated structural porosity, intrinsic mechanical resilience, and distinctive polar chemistry and ion-flux-regulating capability of β -chitin nanofibers. Thus, β -ChNF-S provides a promising and sustainable solution for high-performance Zn-based energy storage technologies.

3.5. Electrochemical performance evaluation of β -ChNF-S

The practical electrochemical performance of β -ChNF-S was further evaluated using full-cell configurations to quantify its impact on energy density, rate capability, and long-term cycling stability. Before performing the galvanostatic analysis, the cyclic voltammetry (CV) profiles of V_2O_5 ||Zn full cells were compared to evaluate the reaction reversibility and kinetic polarization (Fig. S12). Across the tested scan conditions, the β -ChNF-S-based cell exhibited noticeably higher current response and more clearly developed redox features than its GF-S counterpart, indicating improved electrochemical accessibility of the V_2O_5 cathode and enhanced Zn^{2+} insertion/extraction processes through the separator. In particular, the broader current envelope and better-defined anodic/cathodic responses observed with β -ChNF-S suggest reduced overall polarization and more efficient ion-transport pathways in the aqueous environment, consistent with the intrinsically polar nature and electrolyte affinity of the separator. These CV characteristics preliminarily indicate that β -ChNF-S does not merely

separate electrodes physically but modulates ionic transport and interfacial conditions in a full cell, thereby enabling more effective utilization of cathode redox chemistry. Fig. 5a provides an overall comparison of the energy densities achieved with and without considering the separator weight. As shown in the discharge profiles of Fig. 5b, both cells exhibit similar specific capacities of approximately 450 mAh g^{-1} during the first discharge. When only the active material is considered, both systems yield a comparable energy density of about 364 Wh kg^{-1} . However, as illustrated in Fig. 5c, the inclusion of the separator weight reveals a significant advantage for the lightweight β -ChNF-S. The energy density for the β -ChNF-S based cell reaches approximately 80 Wh kg^{-1} , whereas the GF-S based cell drops drastically to about 35 Wh kg^{-1} . This contrast emphasizes that the thinner and lighter nature of β -ChNF-S effectively maximizes the practical energy density of AZIBs by reducing non-active mass. A similar trend is observed in the rate performance tests presented in Fig. 5d and e. While the specific capacities remain comparable at various current densities in Fig. 5d, the energy density considering total separator mass in Fig. 5e is significantly higher for the β -ChNF-S based cell across all tested rates. The β -ChNF-S based cell consistently maintains superior energy density even under high-current conditions where GF-S shows limited efficiency. Cells equipped with β -ChNF-S exhibited excellent capacity retention across all tested current densities, fully recovering their initial capacity when returned to the baseline rate. This behavior demonstrates the outstanding reversibility, rapid ionic conductivity, and structural resilience imparted by the nanofibrous β -ChNF-S architecture. Conversely, cells employing GF-S showed notably inferior rate performance, with incomplete capacity recovery owing to sluggish ionic transport and electrode degradation under elevated current conditions. To elucidate the kinetic origin of the distinct rate reversibility, EIS was performed using $\text{V}_2\text{O}_5|\text{Zn}$ cells (Fig. S13). The Nyquist plots were fitted with an equivalent circuit including solution resistance (R_s), interfacial resistance (R_{int}), charge-transfer resistance (R_{ct}), and a Warburg element (Z_w) associated with mass transport (Barsoukov & Macdonald, 2005). Notably, the β -ChNF-S-

based cell exhibited a substantially smaller semicircle than the GF-S-based cell, corresponding to a markedly reduced fitted R_{ct} ($\sim 450 \Omega$ for β -ChNF-S vs $\sim 1100 \Omega$ for GF-S). This reduced interfacial charge-transfer barrier is consistent with faster reaction kinetics and lower polarization during high-rate operation, both of which suppress the buildup of concentration gradients that can trigger nonuniform Zn deposition and accelerate electrode degradation. Moreover, the inclusion of Z_w in the fitting model indicates that diffusion-related impedance contributes to the overall response, with the β -ChNF-S architecture mitigating mass-transport limitations compared with GF-S owing to its well-wetted polar porous network with continuous ion-transport pathways, particularly under elevated current densities where ionic depletion becomes pronounced. To further monitor the evolution of interfacial resistance during cycling, EIS measurements were additionally performed after the 1st, 10th, and 100th cycles (Fig. S14). Although the GF-S cell exhibited lower R_{ct} at the early stage, its R_{ct} increased markedly after prolonged cycling, reaching 831Ω after 100 cycles. In contrast, the β -ChNF-S cell maintained a more stable R_{ct} evolution and showed a lower R_{ct} of 469Ω after 100 cycles, indicating that β -ChNF-S effectively stabilizes the electrode/electrolyte interface and sustains homogeneous Zn^{2+} flux during cycling. Importantly, the Zn^{2+} transference number was independently evaluated using chronoamperometry combined with EIS before and after polarization in $\text{Zn}||\text{Zn}$ symmetric cells (Fig. S15) (Evans et al., 1987). The β -ChNF-S exhibited a Zn^{2+} transference number comparable to that of GF-S, with values of 0.41 and 0.40, respectively. This result suggests that the enhanced performance of β -ChNF-S is more appropriately attributed to Zn^{2+} flux regulation rather than conventional ion-selective transport. The hydroxyl/acetyl-amide-rich polar framework and excellent electrolyte wettability of β -ChNF-S help maintain continuous ion-transport pathways and homogenize Zn^{2+} distribution near the Zn surface. Together with the reduced interfacial charge-transfer resistance, this flux-regulating behavior contributes to stable Zn plating/stripping and improved high-rate performance. The long-term cycling stability was

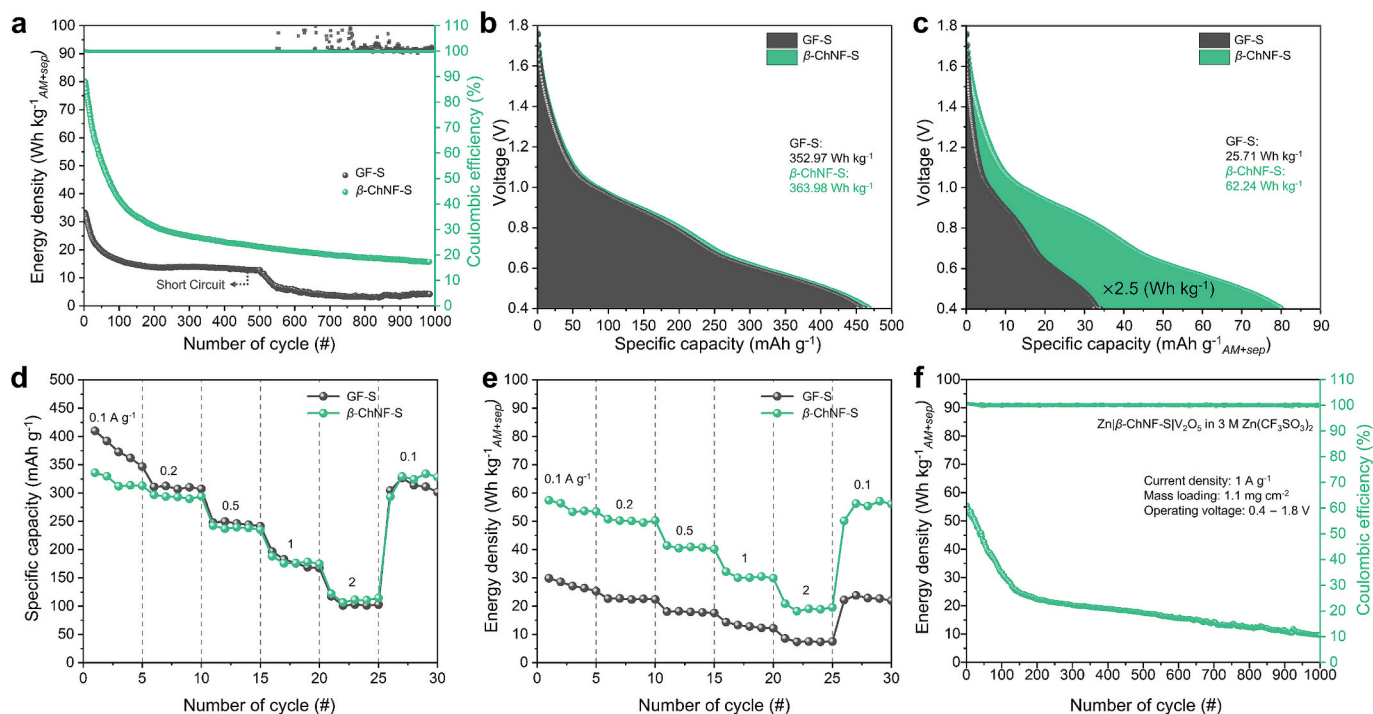


Fig. 5. Electrochemical performance and energy density analysis of $\text{V}_2\text{O}_5|\text{Zn}$ full cells. a) Energy density comparison during cycling considering the separator weight. Discharge voltage profiles based on b) active material mass and c) the specific energy density comparison. Rate capability from 0.1 to 2 A g^{-1} evaluated by d) specific capacity and e) the energy density considering total separator mass (e). (f) Long-term energy density stability and coulombic efficiency of the β -ChNF-S based cell evaluated at 1 A g^{-1} .

further assessed at higher current densities of 1 and 2 A g⁻¹ (Fig. 5f, Fig. S16). The cell demonstrated exceptional durability at this high rate, maintaining a stable energy density and a consistently high Coulombic efficiency approaching 100% throughout the prolonged cycling process. To further clarify the molecular-level interaction responsible for Zn-flux regulation, DFT calculations were conducted for Zn adsorption on β -chitin using both bulk and surface models (Fig. 6a). In the bulk β -chitin configuration, Zn was located in close proximity to the oxygen-containing polar moieties, giving a Zn—O distance of 2.05 Å and an adsorption energy of 2.01 eV. This result confirms that the hydroxyl and acetyl-amide functionalities of β -chitin can provide favorable polar interaction sites for Zn species. By contrast, the β -chitin surface model showed a longer Zn—O distance of 2.38 Å and a substantially lower adsorption energy of 0.10 eV, indicating a weaker and more reversible interaction at the exposed surface. Such moderate surface affinity is beneficial for separator operation because it can guide and redistribute Zn species without causing excessive Zn trapping or sluggish ion transport. Accordingly, these computational results suggest that β -ChNF-S functions as an active ion-regulating medium: its polar functional groups interact with Zn species sufficiently to homogenize Zn²⁺ flux, while the weak surface interaction preserves facile transport across the separator. Combined with the high electrolyte wettability and interconnected fibrous architecture, this molecular-level interaction contributes to uniform Zn deposition and suppressed dendrite formation. Based on these experimental and computational results, the overall synergistic mechanism for the enhanced electrochemical performance of β -ChNF-S is schematically illustrated in Fig. 6b. In contrast to the conventional GF-S, which is inherently thick and heavy, the β -ChNF-S features a significantly thinner and more lightweight architecture. This structural compactness directly contributes to the improved gravimetric (Wh kg⁻¹)

and volumetric (Wh L⁻¹) energy density of the AZIBs. Beyond its physical advantages, the β -ChNF-S acts as an active ion-regulating medium. While the coarse GF-S induces non-uniform Zn²⁺ flux and detrimental vertical dendrite growth, the unique acetyl-amide and hydroxyl chemistry of the β -ChNF-S homogenizes Zn²⁺ transport. This promotes stable, horizontal Zn deposition and effectively suppresses dendrite propagation. Specifically, the β -ChNF-S stabilizes Zn metal cycling through the following synergistic effects:

1. Mitigation of localized side reactions and byproduct accumulation: During Zn plating/stripping in aqueous electrolytes, local pH can increase near the Zn surface due to parasitic reactions such as HER, thereby promoting the precipitation of insulating, insoluble alkaline Zn species depending on electrolyte chemistry (e.g., Zn(OH)₂-like byproducts). The highly hydrophilic β -ChNF-S network, with strong electrolyte affinity and high electrolyte retention, can help mitigate local concentration gradients and interfacial heterogeneity by maintaining continuous electrolyte supply and reducing localized ionic starvation. This moderated interfacial environment is expected to delay the formation of passivating byproducts that induce interfacial heterogeneity and dendrite deposition.
2. Modulation of the interfacial solvation environment and reduced kinetic polarization: The acetyl-amide and hydroxyl groups of β -ChNF-S provide polar and hydrogen-bonding sites that reorganize interfacial water molecules and weakly interact with hydrated Zn species. As supported by electrolyte-wetted FT-IR analysis, the hydroxyl-rich β -ChNF-S framework reconstructs the hydrogen-bonding network of water and reduces weak/free water. Such distributed polarity can modulate the interfacial solvation environment and provide favorable ion-transfer conditions. Consistently, the

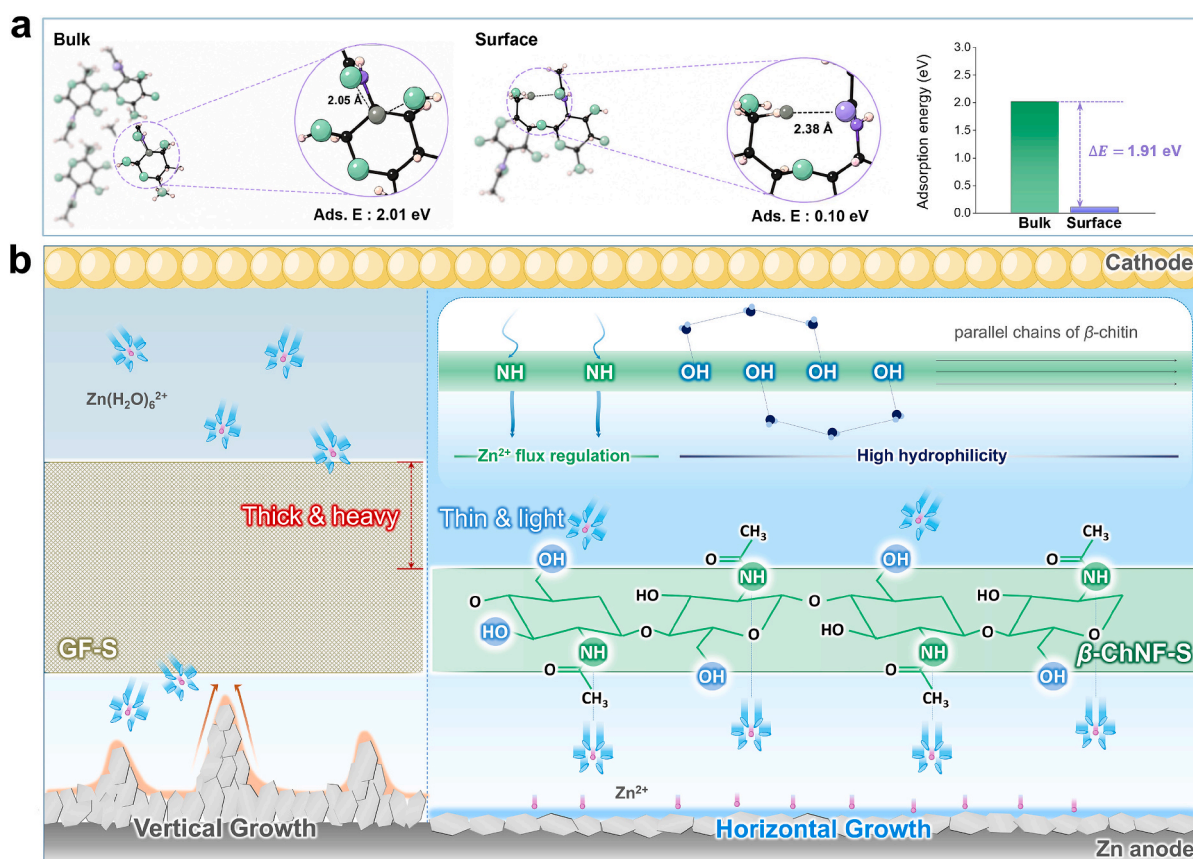


Fig. 6. a) DFT-calculated Zn adsorption behavior on β -chitin bulk and surface models. The Zn—O distance and adsorption energy are 2.05 Å and 2.01 eV for the bulk model, and 2.38 Å and 0.10 eV for the surface model. b) Schematic comparison of Zn²⁺ flux and Zn deposition regulated by the separator. The GF-S induces non-uniform ion flux and dendrite-prone vertical growth, whereas the β -ChNF-S homogenizes Zn²⁺ transport and promotes uniform, dendrite-suppressed deposition.

reduced interfacial polarization (e.g., lower apparent charge-transfer resistance and improved polarization response) indicates that β -ChNF-S lowers the kinetic barriers at the Zn–electrolyte interface, thereby favoring efficient Zn deposition over parasitic pathways.

3. Flux homogenization and suppression of current constriction: The nanofibrous, uniformly wettable architecture of β -ChNF-S promotes homogeneous electrolyte distribution and continuous ion-conduction pathways across the separator thickness. This reduces local current density hotspots and mitigates concentration polarization during high-rate operation. From a classical Sand-type depletion perspective, lowering current constriction delays near-surface cation depletion and thus suppresses dendrite initiation and vertical growth, consistent with the smoother Zn morphology and extended lifetime of symmetric cells assembled with β -ChNF-S (Chazalviel, 1990; Monroe & Newman, 2003).
4. Mechanical/tortuous network blocking of dendrite propagation: Beyond chemistry and transport, β -ChNF-S provides a mechanically continuous, junction-rich fibrous scaffold that increases the tortuosity of potential dendrite penetration pathways. This distributed mechanical resistance helps prevent rapid dendrite piercing and internal short circuits, enhancing prolonged cycling stability under harsh current and areal-capacity conditions. Overall, β -ChNF-S stabilizes Zn metal cycling through the coupled regulation of interfacial polarization, ion-flux homogeneity, and mechanical penetration resistance. Ultimately, chitin nanofibers represent a potential material for high-performance separators because their lightweight and thin nature is exceptionally effective for improving energy density compared to traditional glass fibers.

4. Conclusion

In this work, we developed a sustainable separator for AZIBs, i.e., β -ChNF-S, which was fabricated from squid-pen biomass via centrifugal jet spinning. β -ChNF-S exhibits a mechanically robust, interconnected nanofibrous network that retains crystallinity and abundant polar functionalities of β -chitin. Owing to its high electrolyte affinity, β -ChNF-S exhibits ultrafast wetting, exceptionally high electrolyte uptake, favorable permeability, and excellent thermal stability. These physico-chemical advantages translate into highly uniform Zn deposition with dendrite-suppressed, flat Zn surfaces and markedly extended Zn||Zn symmetric cell stability even under harsh current conditions. In full cells, β -ChNF-S improves practical energy density by reducing separator mass, while delivering enhanced rate capability and long-term cycling durability. Overall, this study demonstrates that bio-derived β -chitin nanofiber membranes can function as ion-regulating separators that simultaneously improve Zn metal stability and cell-performance metrics, providing a scalable and eco-friendly strategy for high-performance AZIBs.

CRedit authorship contribution statement

Jeong-Ho Park: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jin-Hyuk Lee:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis. **Hyeon Min Cho:** Investigation, Formal analysis, Data curation. **Ilguy Kim:** Validation, Investigation. **Ho-Jin Lee:** Investigation, Data curation. **Subin Lee:** Visualization, Investigation. **Dae-Kwon Boo:** Supervision, Resources. **Jae-Hyuk Cho:** Resources, Project administration. **Su Yeon Bae:** Validation, Investigation, Formal analysis. **Daehyun Kim:** Formal analysis, Investigation. **Haesun Park:** Methodology, Supervision. **Jun Young Cheong:** Writing – review & editing, Validation, Investigation. **Byungil Hwang:** Writing – review & editing, Supervision, Resources. **Ki-Ro Yoon:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Jungho Jin:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Ji-Won**

Jung: Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.carbpol.2026.125567>.

Data availability

Data will be made available on request.

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